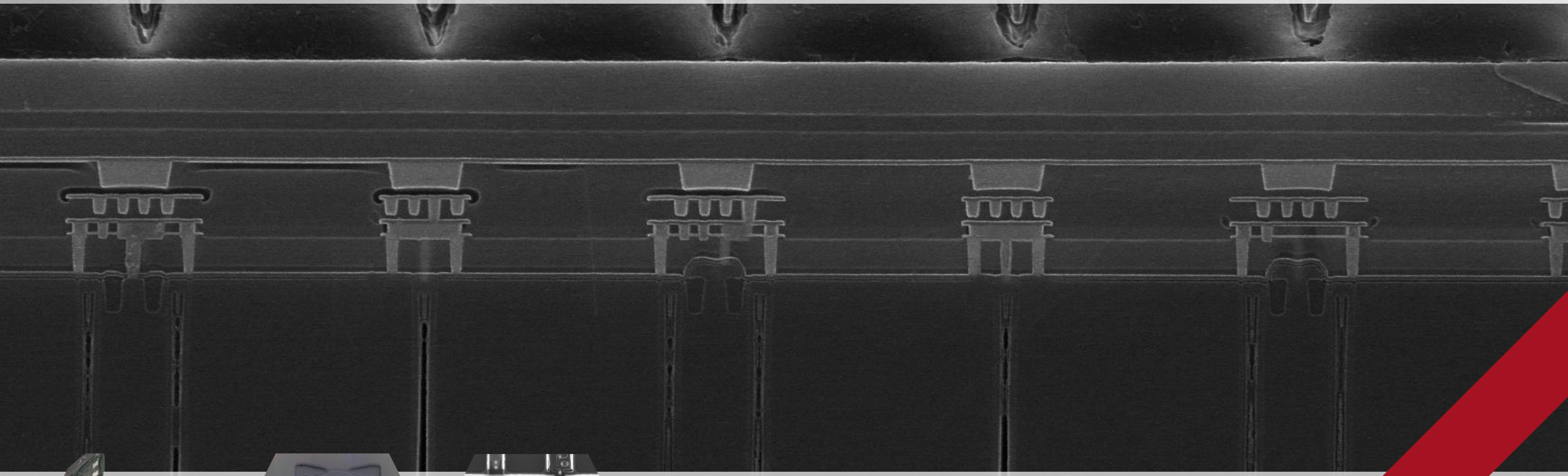
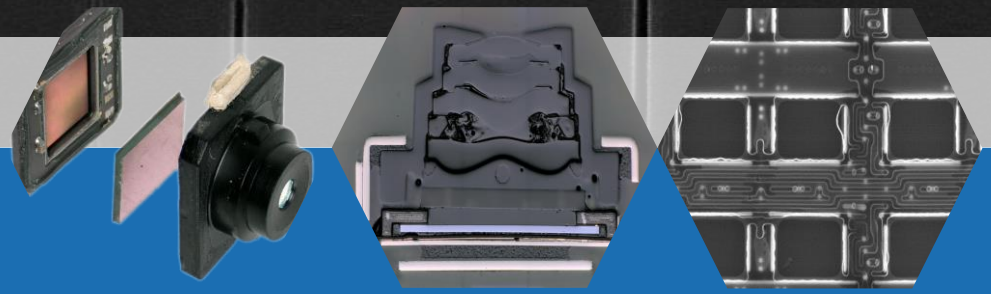
SYSTEMPlus CONSULTING

Reverse Costing & Technology Analysis REPORT





STMicroelectronics NIR Camera Sensor

True Depth Camera Sensor from Apple iPhone X

IMAGING report by Stéphane ELISABETH

December 2017 – version 1

SAMPLE

21 rue la Noue Bras de Fer
44200 NANTES - FRANCE

+33 2 40 18 09 16

info@systemplus.fr

www.systemplus.fr

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- This full reverse costing study has been conducted to provide insight on technology data, manufacturing cost and selling price of the NIR Camera Sensor in the Apple iPhone X supplied by STMicroelectronics.

- Located in the iPhone X's front, around the main speaker, the True Depth system is packaged in one metal enclosure. The system features a dot projector, a red/green/blue camera and a NIR camera sensor. The latter device, provided by STMicroelectronics, is a NIR camera sensor based on an Imager-SOI substrate, involving multiple innovations.

- The SOI substrate allows high quantum efficiency and reduces noises from the substrate. Combined with STMicroelectronics' knowledge of NIR sensing technology, the sensor die has very small pixels, less than 4 µm long, and high resolution of almost two megapixels, thanks to STMicroelectronics' new technology node, reaching below 140 nm on SOI. This allows the sensor to precisely detect faces to unlock the smartphone in a very short response time. The use of deep trenches coupled with SOI substrates provides high dynamic range pixels. This camera assembly uses stud bumping to connect the sensor die in flip-chip configuration, along with an optical module comprising four lenses.

- This report analyzes the complete NIR camera sensor, including a complete analysis of the module and the sensor die, along with a cost analysis and a price estimate for the device. It also includes a physical and technical comparison with other NIR image sensors for 3D sensing, such as those from Infineon, pmd and Tower Semiconductor for consumer applications, Melexis for automotive, and Texas Instruments for industrial applications.

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Apple iPhone X Teardown

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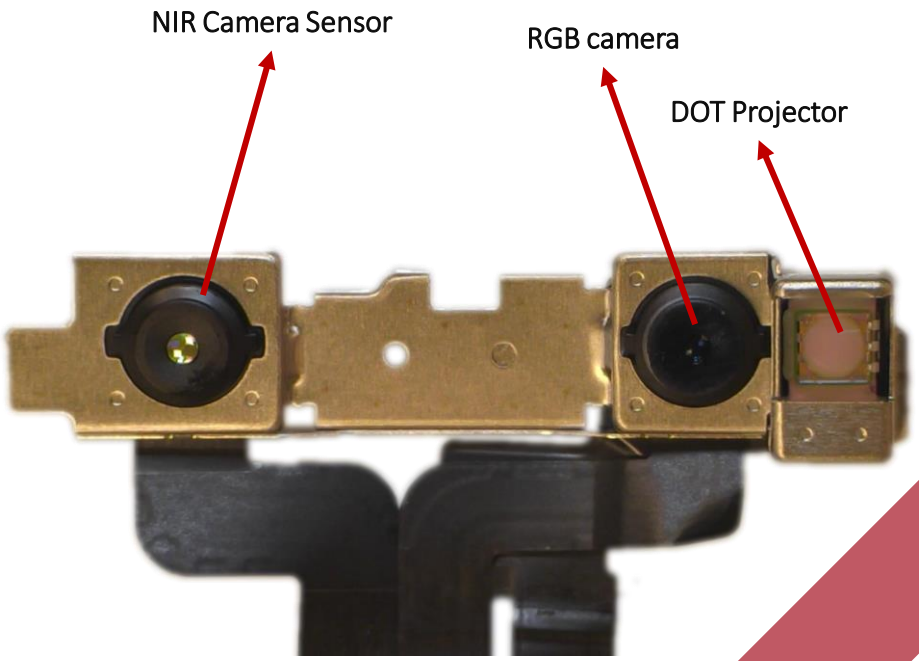
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Apple iPhone X Opened View

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TrueDepth Module

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TrueDepth Module

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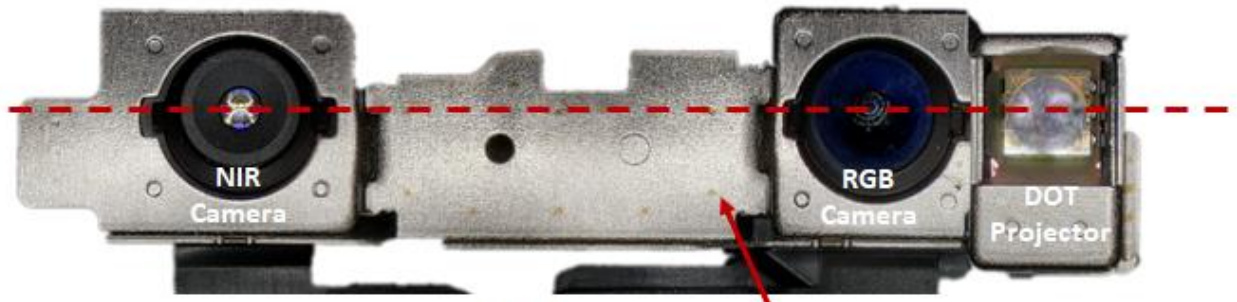
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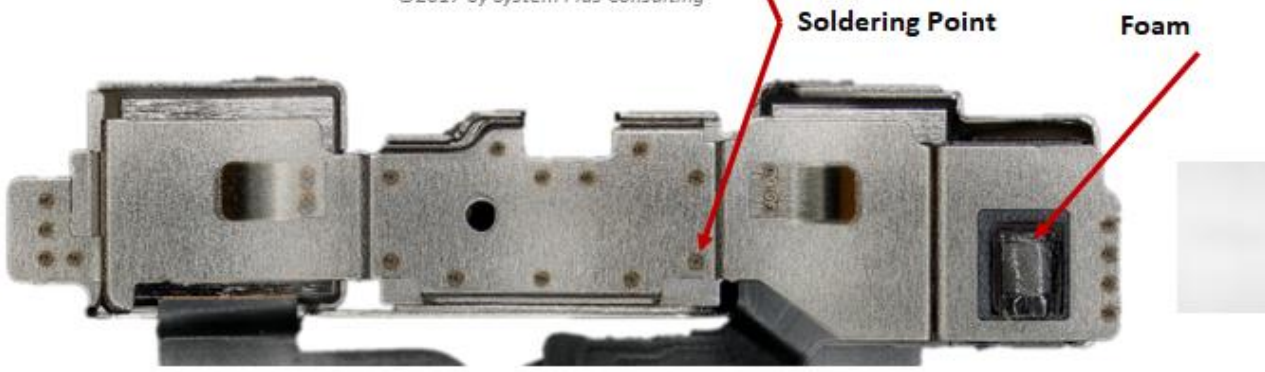
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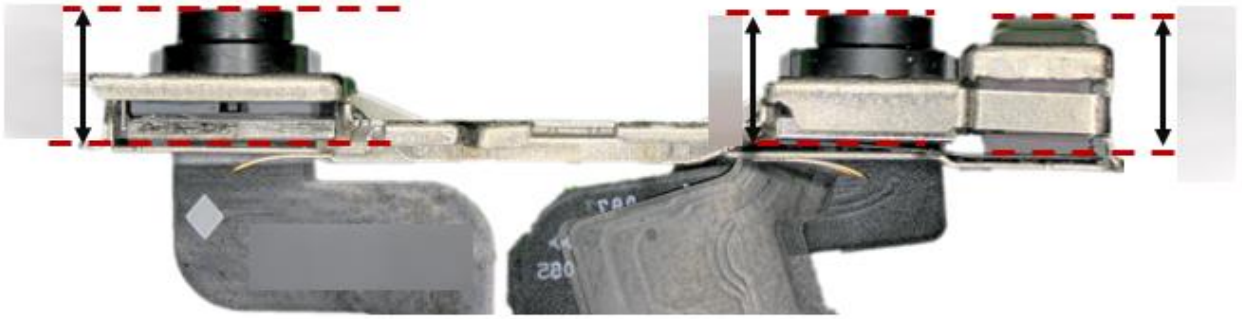
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TrueDepth Module – Top View
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TrueDepth Module – Bottom View
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TrueDepth Module – Side View
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Module Views & Dimensions

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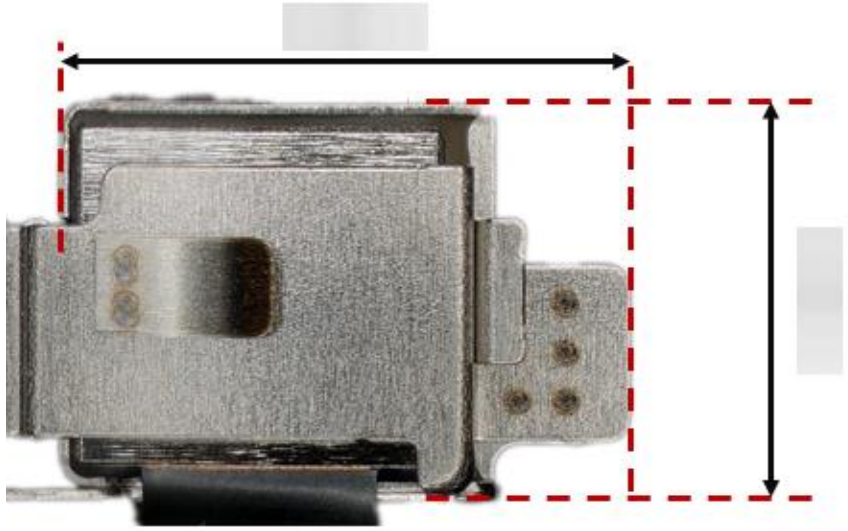
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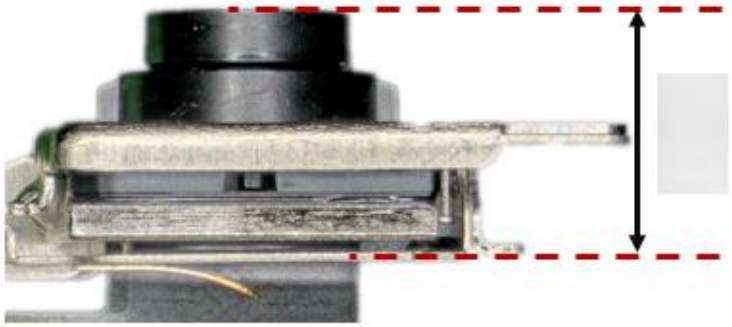
NIR Camera Sensor Bottom View
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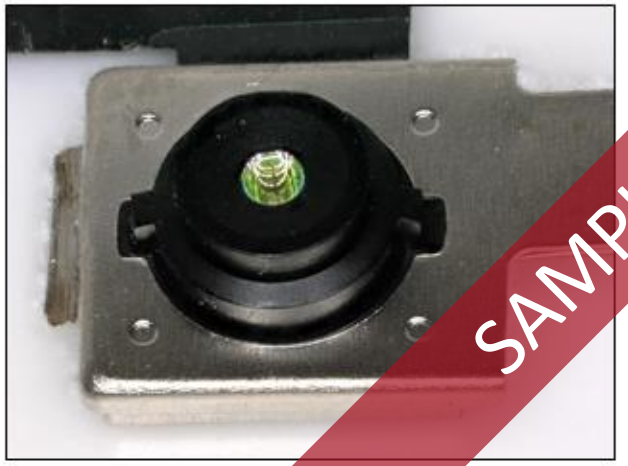
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NIR Camera Sensor – Flex Marking
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NIR Camera Sensor Global View
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Module Disassembly

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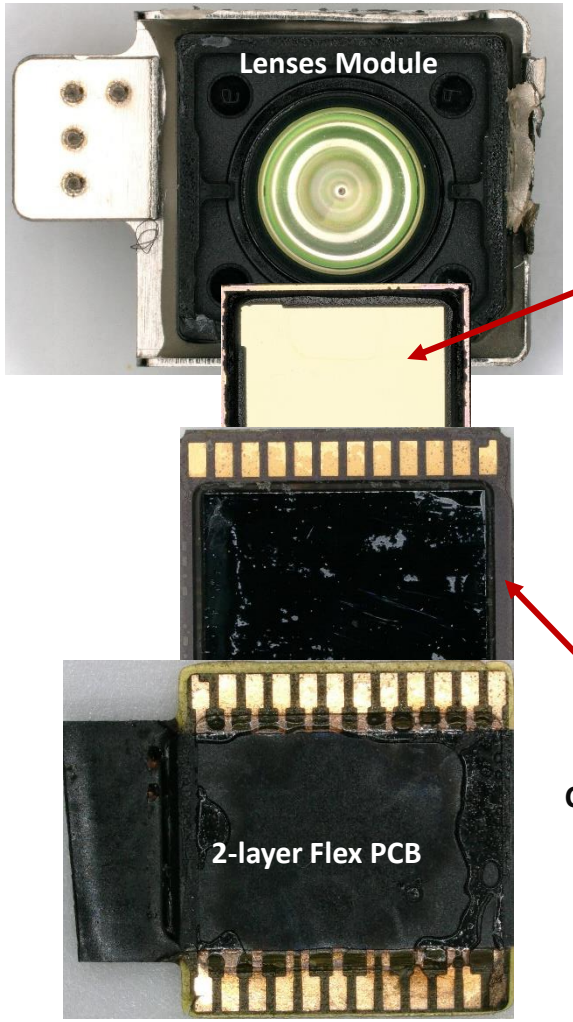
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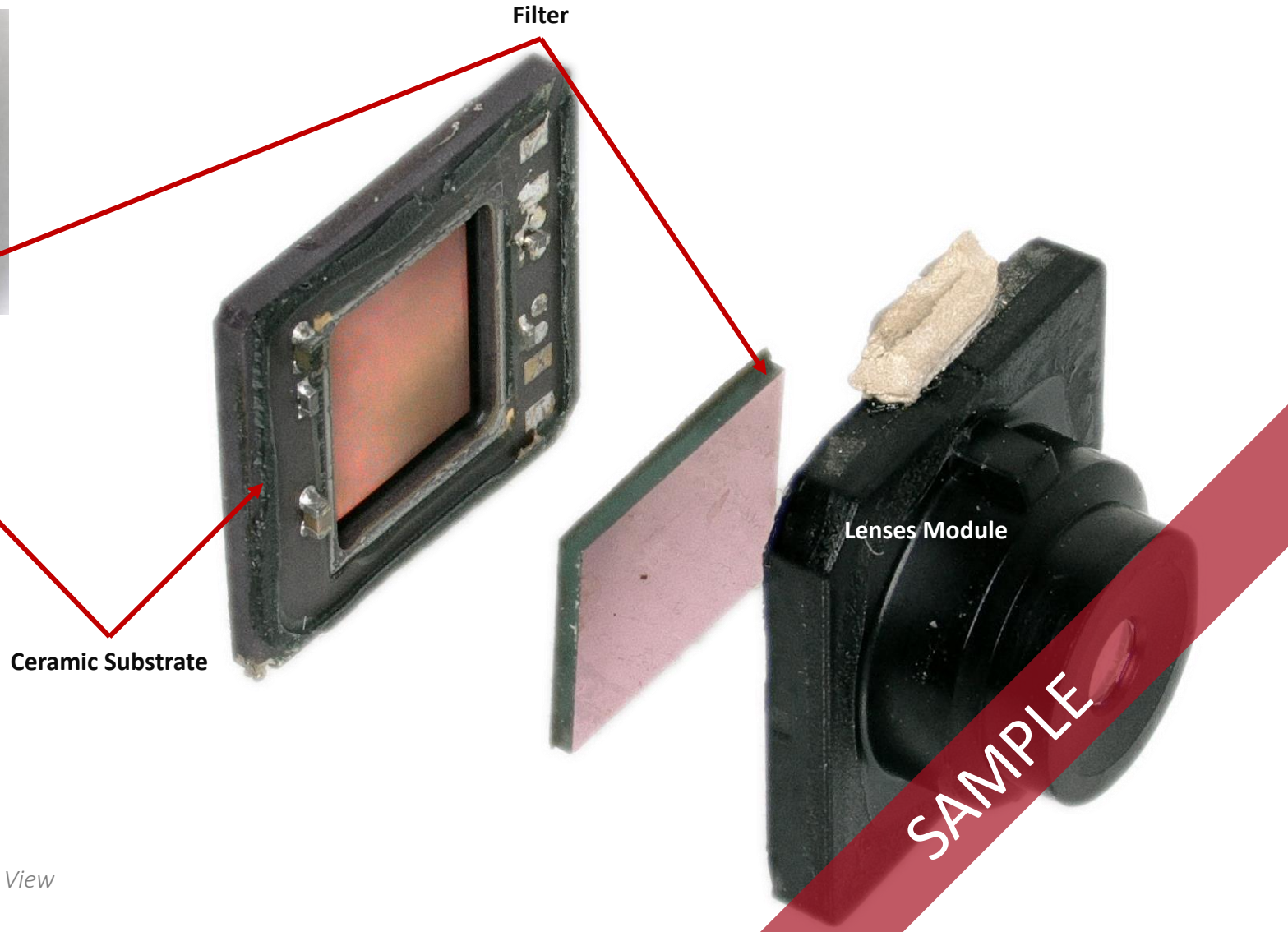
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NIR Camera Sensor – Disassembly – Bottom View

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NIR Camera Sensor – Disassembly – Global View

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Module Cross-Section

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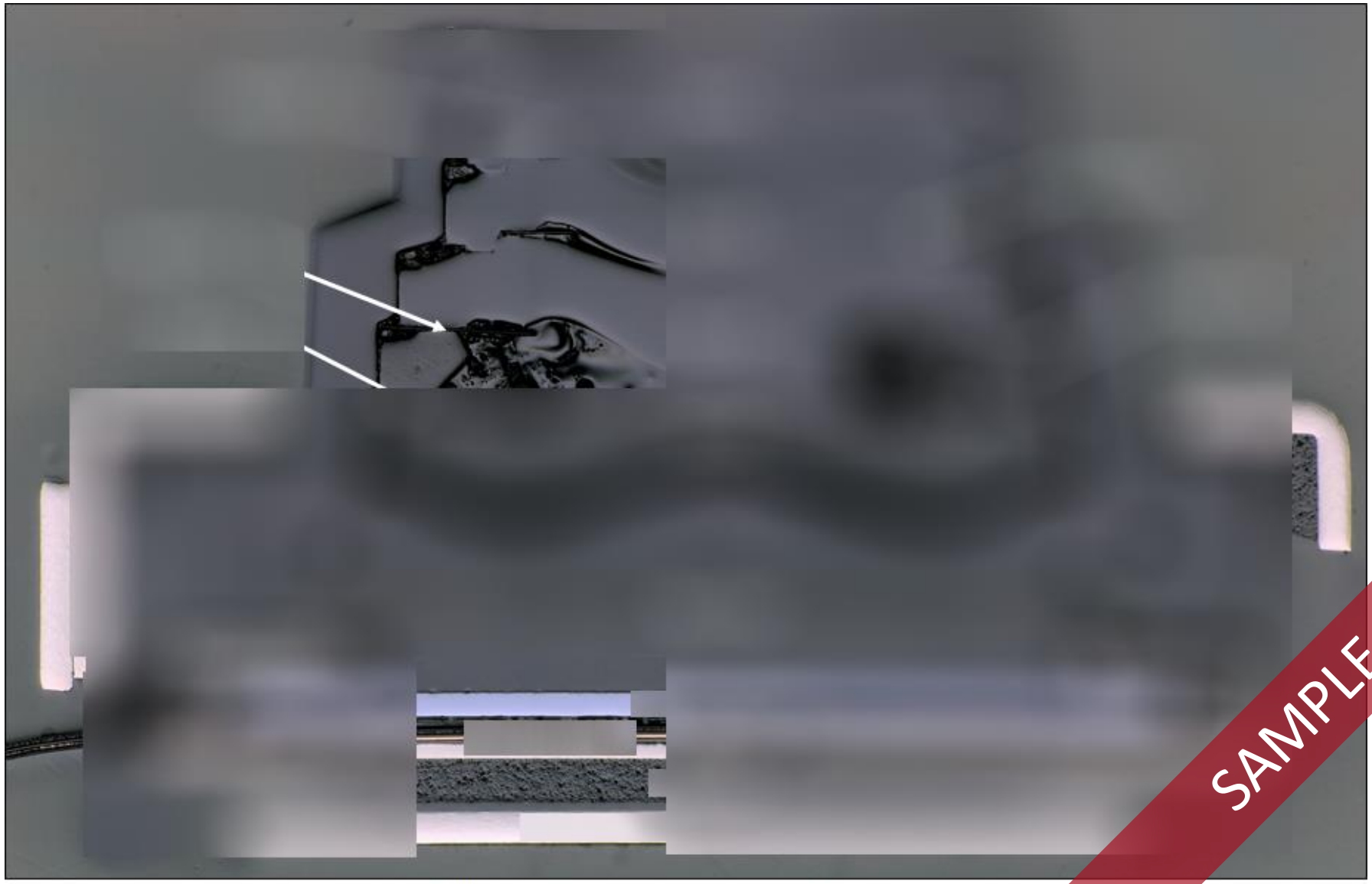
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NIR Camera Sensor – Cross-Section – Optical View

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Module Cross-Section – Filter

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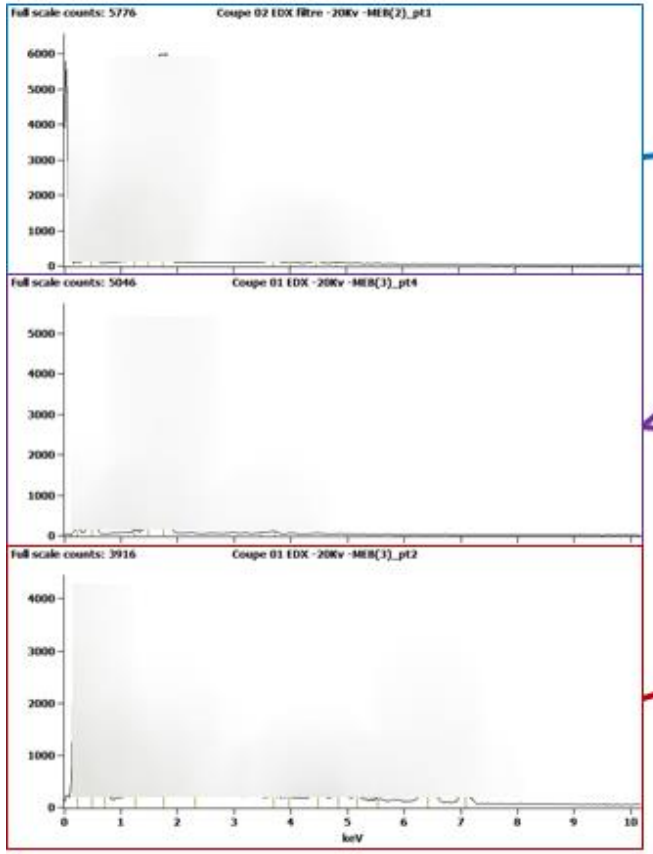
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NIR Camera Sensor – Cross-Section

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NIR Camera Sensor – Filter Cross-Section – SEM View

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Sensor Die – Die View & Dimensions

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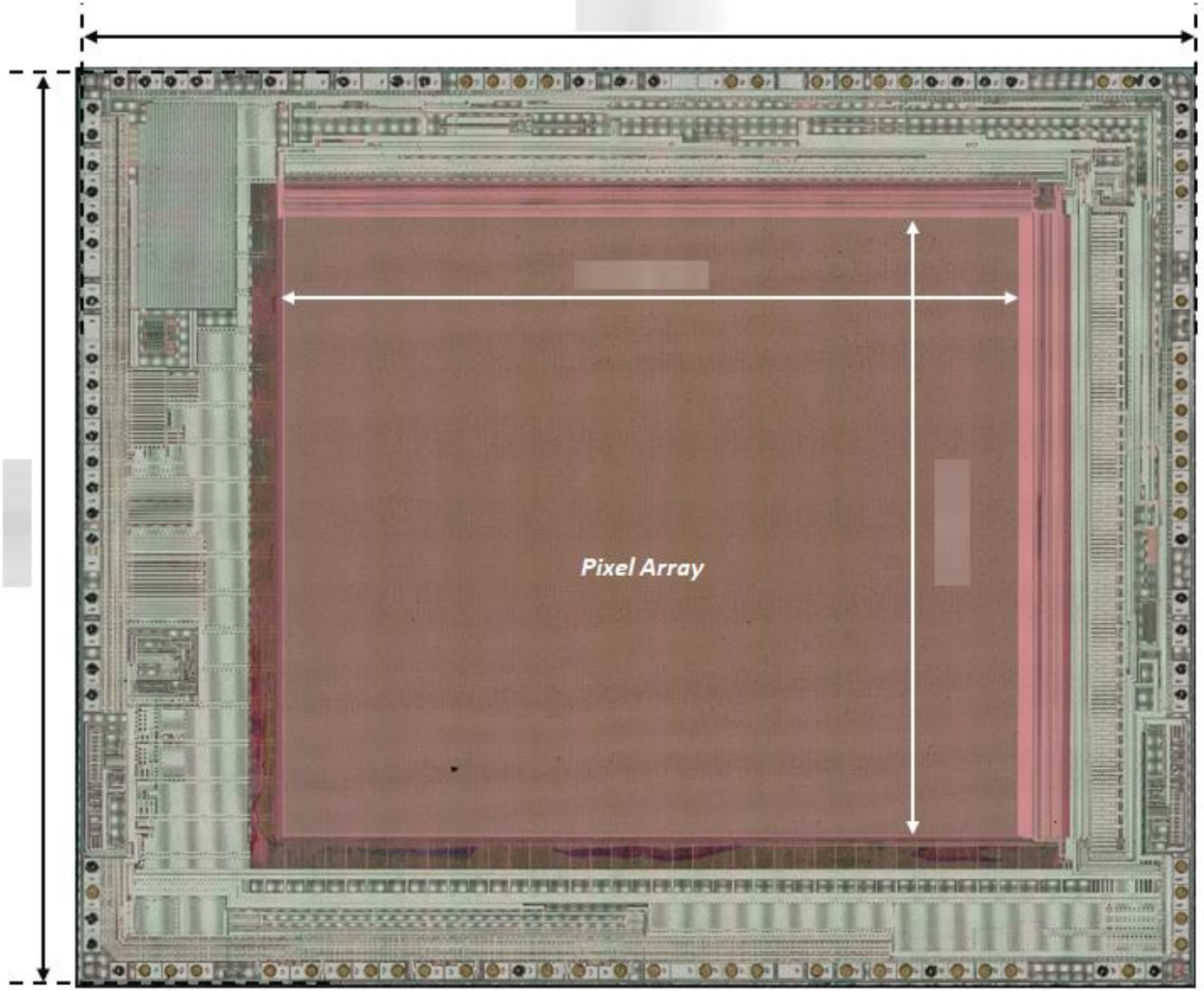
Manufacturing Process Flow

Cost Analysis

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Die Top View – Optical View

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Die Area:

Nb of PGDW per 12-inch wafer:

Pad number:
- Connected:

Pixel Array:

NIR Image Sensor resolution:
- Pixel Area:
- Pixel Size:

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Sensor Die – Die Delayering – Pixels

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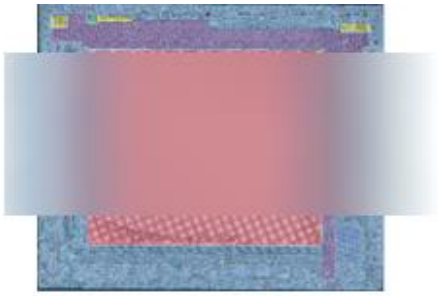
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Die Delayering – Optical View

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-
- Pixel Area:
- Photodiode Area:
- Fill Factor:



Sensor Die – Pixels Deprocessing – SEM View

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Sensor Die – Die Cross-Section – Pixels

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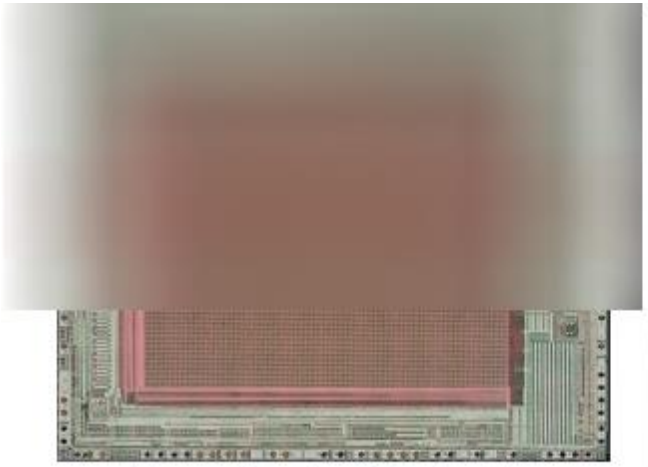
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Sensor Die – Cross-Section Plan

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Sensor Die – Die Cross-Section – Pixels – SEM View

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Sensor Die – Die Cross-Section – Pixels

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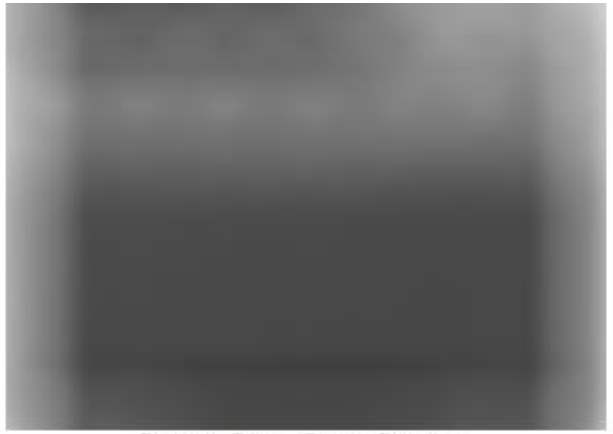
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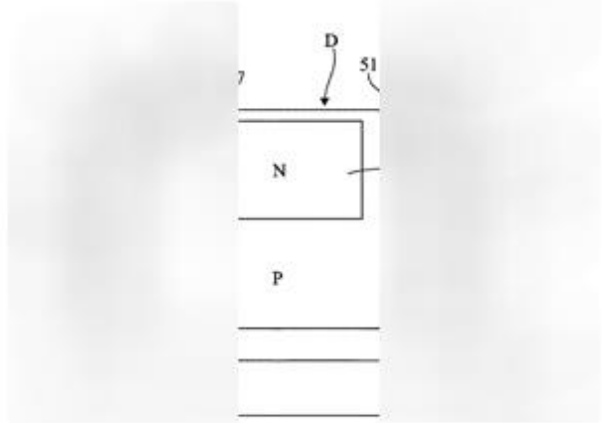
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Sensor Die – Cross-Section
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Sensor Die – Die Cross-Section – Pixels – Schematic View



Sensor Die – Die Cross-Section – Pixels After Doping Revelation – SEM View
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Estimated Pixel Performance

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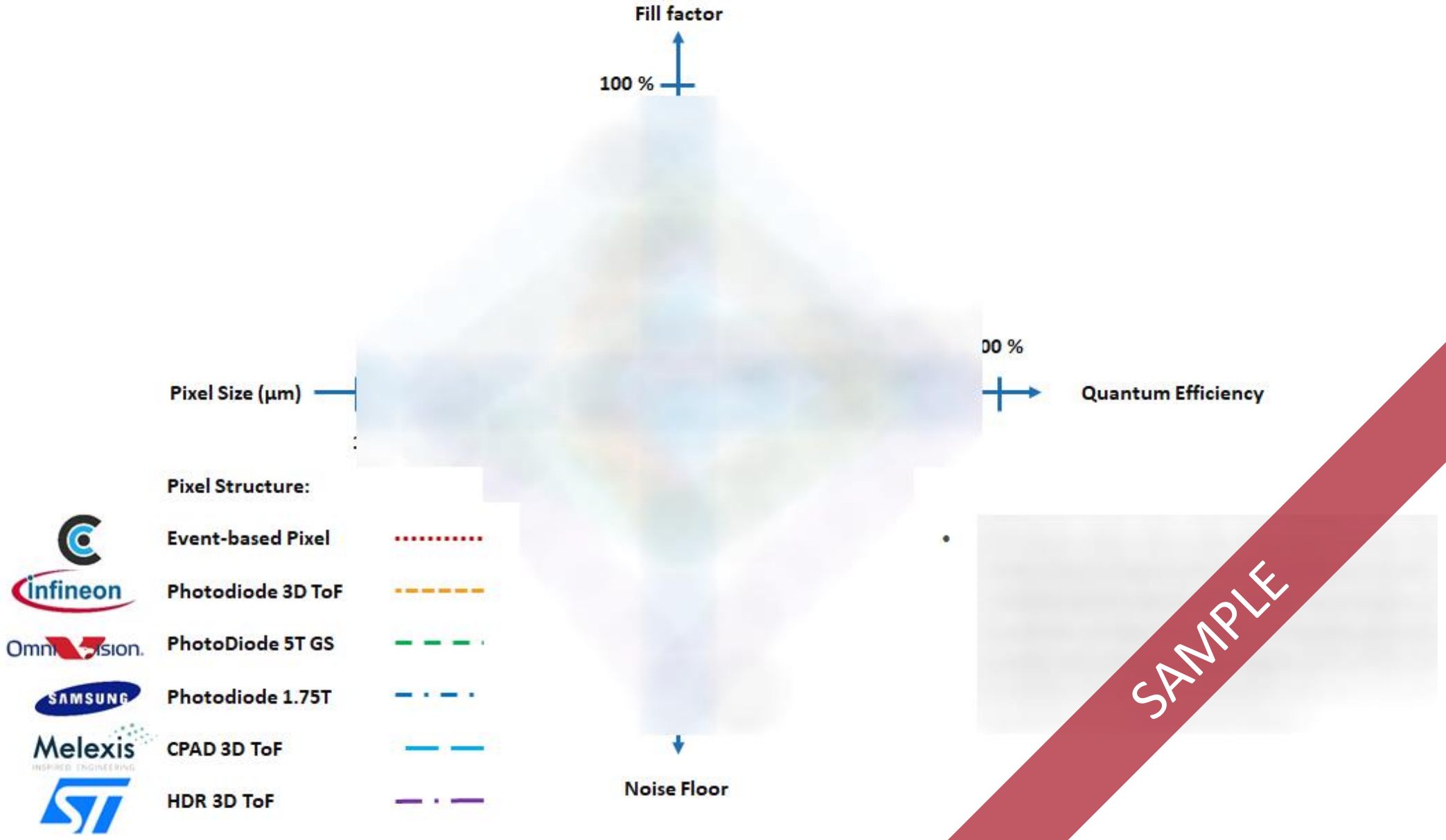
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Pixel Array Circuit Process Flow (1/2)

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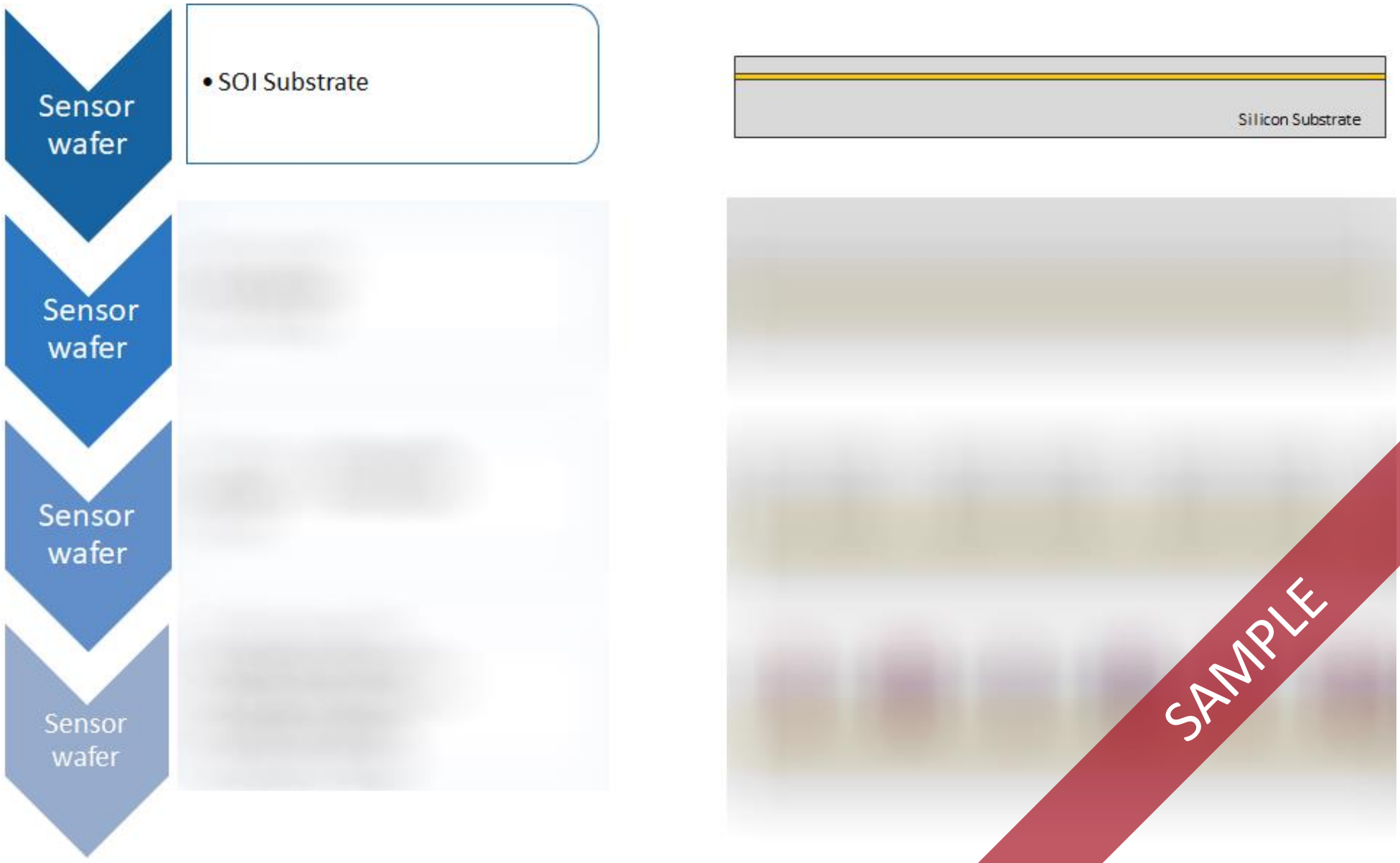
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Sensor Front-End Cost

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Front-End	Low Yield		Medium Yield		High Yield	
	Cost	Breakdown	Cost	Breakdown	Cost	Breakdown
Raw Wafer Cost (SOI)						
Clean Room Cost						
Equipment Cost						
Consumable Cost						
Labor Cost						
Yield losses Cost						
Sensor Front-End Cost						

Sensor Front-End Cost Breakdown (Medium Yield)

The front-end cost for the Sensor ranges from [redacted] according to yield variations.

The largest portion of the manufacturing cost is due to the [redacted]

SAMPLE

Total Front-End Cost

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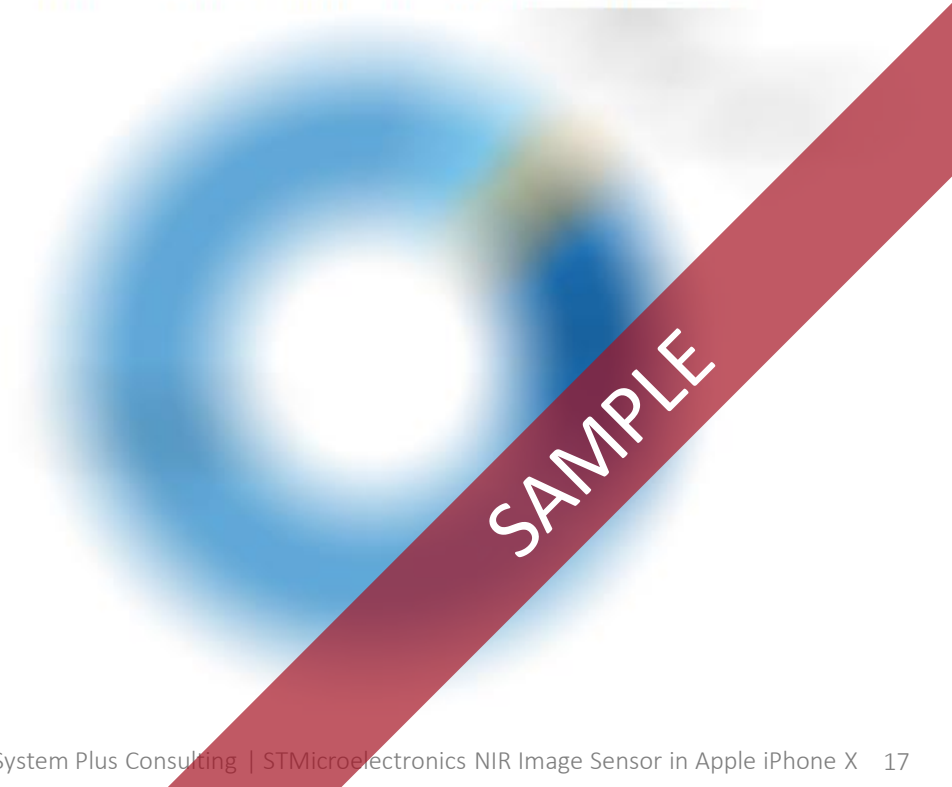
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Total Front-End	Low Yield		Medium Yield		High Yield	
	Cost	Breakdown	Cost	Breakdown	Cost	Breakdown
Raw Substrate (SOI)						
NIR Image Sensor						
Microlenses						
Yield Losses Cost						
Total Front-End Cost						

Total Front-End Cost Breakdown (Medium Yield)



The Total front-end cost for the NIR Image sensor ranges from [redacted] according to yield variations.

The largest portion of the manufacturing cost is due to the [redacted]

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Sensor Wafer & Die Sensor

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	Low Yield		Medium Yield		High Yield	
	Cost	Breakdown	Cost	Breakdown	Cost	Breakdown
Front-End Cost						
BE : Probe Test Cost						
BE : Backgrinding & Dicing Cost						
Total Wafer Cost						
Nb of potential dies per wafer						
Nb of good dies per wafer						
Front-End Cost						
BE : Probe Test, Backgrinding & Dicing Cost						
BE : Yield losses						
Die Cost						

By adding the probe test cost and the dicing, the **Sensor wafer cost** ranges from according to yield variations.

The number of **good dies per wafer** is estimated to ranges from according to yield variations which results in a **die cost** ranging from

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NIR Image Sensor Estimated Manufacturer Price

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	Low Yield		Medium Yield		High Yield	
	Cost	Breakdown	Cost	Breakdown	Cost	Breakdown
Component cost						
Component price						

We estimate that STMicroelectronics realizes a gross margin on the NIR Image Sensor, which results in a final component price ranging from

This corresponds to the selling price for large volume to OEMs.



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Related Reports

REVERSE COSTING ANALYSES - SYSTEM PLUS CONSULTING

IMAGING

- Lenovo Phab2Pro 3D Time of Flight (ToF) Camera
- Apple iPhone X – Infrared Dot Projector
- STMicroelectronics Time of Flight Proximity Sensor in the Apple iPhone 7 Plus

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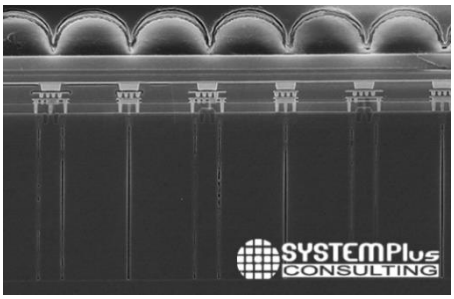
IMAGING

- 3D Imaging and Sensing 2017

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STMicroelectronics' Near Infrared Camera Sensor in the Apple iPhone X

The first NIR camera sensor with multiple innovations based on imager-silicon-on-insulator substrate from SOITEC, supplied and produced by STMicroelectronics for the Apple True Depth Module



Ahead of all its competitors, Apple has started using 3D sensing. For the iPhone's 10th anniversary, it integrated this new functionality, under the name of 'True Depth sensing'. Rather than scene sensing, like competitors such as Lenovo, the 3D Time of Flight system provided by STMicroelectronics and ams enables face recognition and targets security applications. Working along with a dot projector, the near infrared (NIR) camera sensor allows high-precision depth sensing. This NIR camera sensor is the first to feature Imager-Substrate On Insulator (SOI) technology provided by SOITEC, allowing higher quantum efficiency with very low noise.

Located in the iPhone X's front, around the main speaker, the True Depth system is packaged in one metal enclosure. The system features a dot projector, a red/green/blue camera and a NIR camera sensor. The latter device, provided by STMicroelectronics, is a NIR camera sensor based on an Imager-SOI substrate, involving multiple innovations.

The SOI substrate allows high quantum efficiency and reduces noises from the substrate. Combined with STMicroelectronics' knowledge of NIR sensing technology, the sensor die has very small pixels, less than 4 µm long, and high resolution of almost two megapixels, thanks to STMicroelectronics' new technology node, reaching below 140 nm on SOI. This allows the sensor to precisely detect faces to unlock the smartphone in a very short response time. The use of deep trenches coupled with SOI substrates provides high dynamic range pixels. This camera assembly uses stud bumping to connect the sensor die in flip-chip configuration, along with an optical module comprising four lenses.

This report analyzes the complete NIR camera sensor, including a full analysis of the module and the sensor die, along with a cost analysis and a price estimate for the device. It also includes a physical and technical comparison with other NIR image sensors for 3D sensing, such as those from Infineon, pmd and Tower Semiconductor for consumer applications, Melexis for automotive, and Texas Instruments for industrial applications.

Title: STMicroelectronics NIR Camera Sensor

Pages: 97

Date: December 2017

Format: PDF & Excel file

Price: Full report: EUR 3,490

COMPLETE TEARDOWN WITH:

- Detailed photos
- Precise measurements
- Materials analysis
- Manufacturing process flow
- Supply chain evaluation
- Manufacturing cost analysis
- Estimated sales price

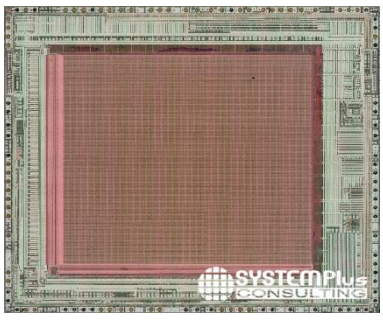




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Estimated Price Analysis

AUTHORS:



Dr. Stéphane Elisabeth

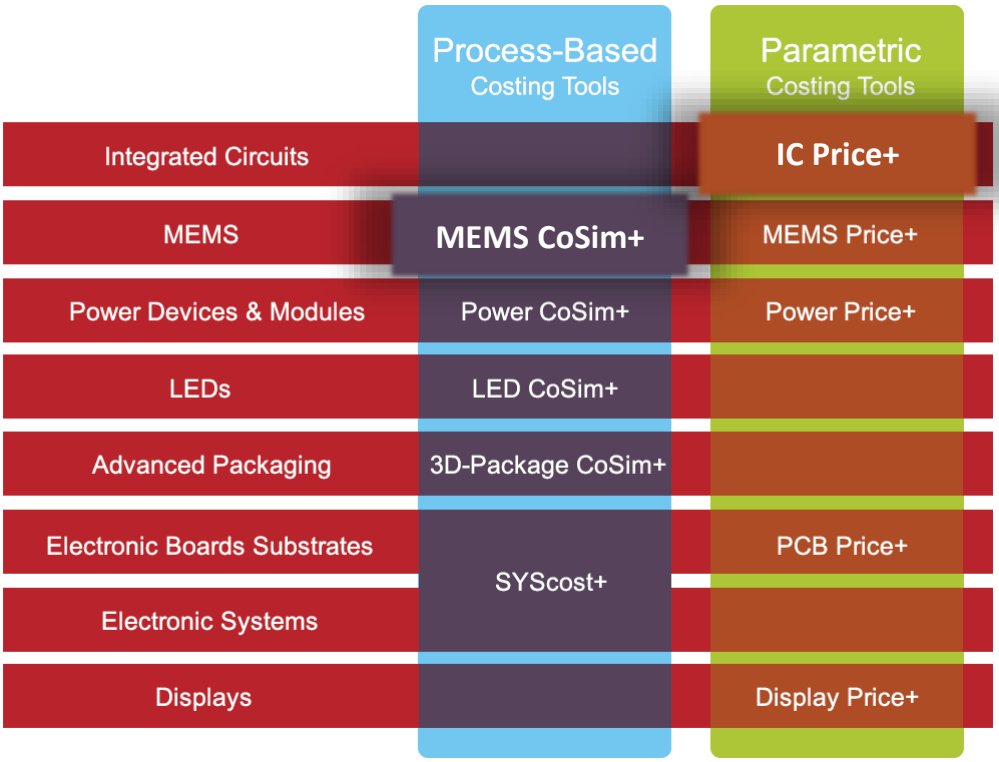
Stéphane has a deep knowledge of materials characterizations and electronics systems. He holds an Engineering Degree in Electronics and Numerical Technology, and a PhD in Materials for Microelectronics.



Nicolas Radufe (Lab)

Nicolas is in charge of physical analysis. He has a deep knowledge in chemical and physical analyses. He previously worked in microelectronics R&D for CEA/LETI in Grenoble and for STMicroelectronics in Crolles.

ANALYSIS PERFORMED WITH OUR COSTING TOOLS MEMS COSIM+ AND IC PRICE+



System Plus Consulting offers powerful costing tools to evaluate the production cost and selling price from single chip to complex structures.

MEMS CoSim+

Cost simulation tool to evaluate the cost of any MEMS process or device.

IC Price+

The tool performs the necessary cost simulation of any Integrated Circuit: ASICs, microcontrollers, DSP, memories, smartpower, and more.

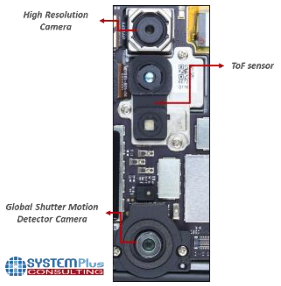
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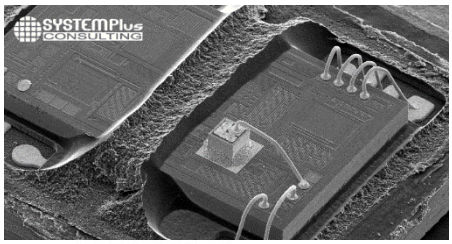
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- Greater China: Mavis - Wang@yole.fr
- Asia: Takashi - Onozawa@yole.fr
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“Seller”: Based in Lyon (France headquarters), Yole Développement is a market research and business development consultancy company, facilitating market access for advanced technology industrial projects. With more than 20 market analysts, Yole works worldwide with the key industrial companies, R&D institutes and investors to help them understand the markets and technology trends.

1. Scope

1.1 The Contracting Parties undertake to observe the following general conditions when agreed by the Buyer and the Seller. ANY ADDITIONAL, DIFFERENT, OR CONFLICTING TERMS AND CONDITIONS IN ANY OTHER DOCUMENTS ISSUED BY THE BUYER AT ANY TIME ARE HEREBY OBJECTED TO BY THE SELLER, SHALL BE WHOLLY INAPPLICABLE TO ANY SALE MADE HEREUNDER AND SHALL NOT BE BINDING IN ANY WAY ON THE SELLER.
1.2 This agreement becomes valid and enforceable between the Contracting Parties after clear and non-equivocal consent by any duly authorized person representing the Buyer. For these purposes, the Buyer accepts these conditions of sales when signing the purchase order which mentions “I hereby accept Yole’s Terms and Conditions of Sale”. This results in acceptance by the Buyer.
1.3 Orders are deemed to be accepted only upon written acceptance and confirmation by the Seller, within [7 days] from the date of order, to be sent either by email or to the Buyer’s address. In the absence of any confirmation in writing, orders shall be deemed to have been accepted.

2. Mailing of the Products

2.1 Products are sent by email to the Buyer:
- within [1] month from the order for Products already released; or
- within a reasonable time for Products ordered prior to their effective release. In this case, the Seller shall use its best endeavours to inform the Buyer of an indicative release date and the evolution of the work in progress.

2.2 Some weeks prior to the release date the Seller can propose a pre-release discount to the Buyer
The Seller shall by no means be responsible for any delay in respect of article 2.2 above, and including incases where a new event or access to new contradictory information would require for the analyst extra time to compute or compare the data in order to enable the Seller to deliver a high quality Products.
2.3 The mailing of the Product will occur only upon payment by the Buyer, in accordance with the conditions contained in article 3.
2.4. The mailing is operated through electronic means either by email via the sales department or automatically online via an email/password. If the Product’s electronic delivery format is defective, the Seller undertakes to replace it at no charge to the Buyer provided that it is informed of the defective formatting within 90 days from the date of the original download or receipt of the Product.
2.5 The person receiving the Products on behalf of the Buyer shall immediately verify the quality of the Products and their conformity to the order. Any claim for apparent defects or for non-conformity shall be sent in writing to the Seller within 8 days of receipt of the Products. For this purpose, the Buyer agrees to produce sufficient evidence of such defects. .
2.6 No return of Products shall be accepted without prior information to the Seller, even in case of delayed delivery. Any Product returned to the Seller without providing prior information to the Seller as required under article 2.5 shall remain at the Buyer’s risk.

3. Price, invoicing and payment

3.1 Prices are given in the orders corresponding to each Product sold on a unit basis or corresponding to annual subscriptions. They are expressed to be inclusive of all taxes. The prices may be reevaluated from time to time. The effective price is deemed to be the one applicable at the time of the order.
3.2 Yole may offer a pre release discount for the companies willing to acquire in the future the specific report and agreeing on the fact that the report may be release later than the anticipated release date. In exchange to this uncertainty, the company will get a discount that can vary from 15% to 10%.
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Branch code: 00170
Account n°: 0170 200 1565 87
BIC or SWIFT code: CCFRFRPP
IBAN: FR76 3005 6001 7001 7020 0156 587
To ensure the payments, the Seller reserves the right to request down payments from the Buyer. In this case, the need of down payments will be mentioned on the order.
3.4 Payment is due by the Buyer to the Seller within 30 days from invoice date, except in the case of a particular written agreement. If the Buyer fails to pay within this time and fails to contact the Seller, the latter shall be entitled to invoice interest in arrears based on the annual rate Refi of the «BCE» + 7 points, in accordance with article L. 441-6 of the French Commercial Code. Our publications (report, database, tool...) are delivered only after reception of the payment.
3.5 In the event of termination of the contract, or of misconduct, during the contract, the Seller will have the right to invoice at the stage in progress, and to take legal action for damages.

4. Liabilities

4.1 The Buyer or any other individual or legal person acting on its behalf, being a business user buying the Products for its business activities, shall be solely responsible for choosing the Products and for the use and interpretations he makes of the documents it purchases, of the results he obtains, and of the advice and acts it deduces thereof.
4.2 The Seller shall only be liable for (i) direct and (ii) foreseeable pecuniary loss, caused by the Products or arising from a material breach of this agreement
4.3 In no event shall the Seller be liable for:
a) damages of any kind, including without limitation, incidental or consequential damages (including, but not limited to, damages for loss of profits, business interruption and loss of programs or information) arising out of the use of or inability to use the Seller’s website or the Products, or any information provided on the website, or in the Products;
b) any claim attributable to errors, omissions or other inaccuracies in the Product or interpretations thereof.
4.4All the information contained in the Products has been obtained from sources believed to be reliable. The Seller does not warrant the accuracy, completeness adequacy or reliability of such information, which cannot be guaranteed to be free from errors.
4.5 All the Products that the Seller sells may, upon prior notice to the Buyer from time to time be modified by or substituted with similar Products meeting the needs of the Buyer. This modification shall not lead to the liability of the Seller, provided that the Seller ensures the substituted Product is similar to the Product initially ordered.
4.6 In the case where, after inspection, it is acknowledged that the Products contain defects, the Seller undertakes to replace the defective products as far as the supplies allow and without indemnities or compensation of any kind for labor costs, delays, loss caused or any other reason. The replacement is guaranteed for a maximum of two months starting from the delivery date. Any replacement is excluded for any event as set out in article 5 below.
4.7 The deadlines that the Seller is asked to state for the mailing of the Products are given for information only and are not guaranteed. If such deadlines are not met, it shall not lead to any damages or cancellation of the orders, except for non acceptable delays exceeding [4] months from the stated deadline, without information from the Seller. In such case only, the Buyer shall be entitled to ask for a reimbursement of its first down payment to the exclusion of any further damages.
4.8 The Seller does not make any warranties, express or implied, including, without limitation, those of sale ability and fitness for a particular purpose, with respect to the Products. Although the Seller shall take reasonable steps to screen Products for infection of viruses, worms, Trojan horses or other codes containing contaminating or destructive properties before making the Products available, the Seller cannot guarantee that any Product will be free from infection.

5. Force majeure

The Seller shall not be liable for any delay in performance directly or indirectly caused by or resulting from acts of nature, fire, flood, accident, riot, war, government intervention, embargoes, strikes, labor difficulties, equipment failure, late deliveries by suppliers or other difficulties which are beyond the control, and not the fault of the Seller.

6. Protection of the Seller’s IPR

6.1 All the IPR attached to the Products are and remain the property of the Seller and are protected under French and international copyright law and conventions.
6.2 The Buyer agreed not to disclose, copy, reproduce, redistribute, resell or publish the Product, or any part of it to any other party other than employees of its company. The Buyer shall have the right to use the Products solely for its own internal information purposes. In particular, the Buyer shall therefore not use the Product for purposes such as:
- Information storage and retrieval systems;
- Recordings and re-transmittals over any network (including any local area network);
- Use in any timesharing, service bureau, bulletin board or similar arrangement or public display;
- Posting any Product to any other online service (including bulletin boards or the Internet);
- Licensing, leasing, selling, offering for sale or assigning the Product.

6.3 The Buyer shall be solely responsible towards the Seller of all infringements of this obligation, whether this infringement comes from its employees or any person to whom the Buyer has sent the Products and shall personally take care of any related proceedings, and the Buyer shall bear related financial consequences in their entirety.
6.4 The Buyer shall define within its company point of contact for the needs of the contract. This person will be the recipient of each new report in PDF format. This person shall also be responsible for respect of the copyrights and will guaranty that the Products are not disseminated out of the company.
6.5 In the context of annual subscriptions, the person of contact shall decide who within the Buyer, shall be entitled to access on line the reports on I-micronews.com. In this respect, the Seller will give the Buyer a maximum of 10 password, unless the multiple sites organization of the Buyer requires more passwords. The Seller reserves the right to check from time to time the correct use of this password.
6.6 In the case of a multisite, multi license, only the employee of the buyer can access the report or the employee of the companies in which the buyer have 100% shares. As a matter of fact the investor of a company, the joint venture done with a third party etc..cannot access the report and should pay a full license price.

7. Termination

7.1 If the Buyer cancels the order in whole or in part or postpones the date of mailing, the Buyer shall indemnify the Seller for the entire costs that have been incurred as at the date of notification by the Buyer of such delay or cancellation. This may also apply for any other direct or indirect consequential loss that may be borne by the Seller, following this decision.
7.2 In the event of breach by one Party under these conditions or the order, the non-breaching Party may send a notification to the other by recorded delivery letter upon which, after a period of thirty (30) days without solving the problem, the non-breaching Party shall be entitled to terminate all the pending orders, without being liable for any compensation.

8. Miscellaneous

All the provisions of these Terms and Conditions are for the benefit of the Seller itself, but also for its licensors, employees and agents. Each of them is entitled to assert and enforce those provisions against the Buyer.
Any notices under these Terms and Conditions shall be given in writing. They shall be effective upon receipt by the other Party.
The Seller may, from time to time, update these Terms and Conditions and the Buyer, is deemed to have accepted the latest version of these terms and conditions, provided they have been communicated to him in due time.

9. Governing law and jurisdiction

9.1 Any dispute arising out or linked to these Terms and Conditions or to any contract (orders) entered into in application of these Terms and Conditions shall be settled by the French Commercial Courts of Lyon, which shall have exclusive jurisdiction upon such issues.

9.2 French law shall govern the relation between the Buyer and the Seller, in accordance with these Terms and Conditions.

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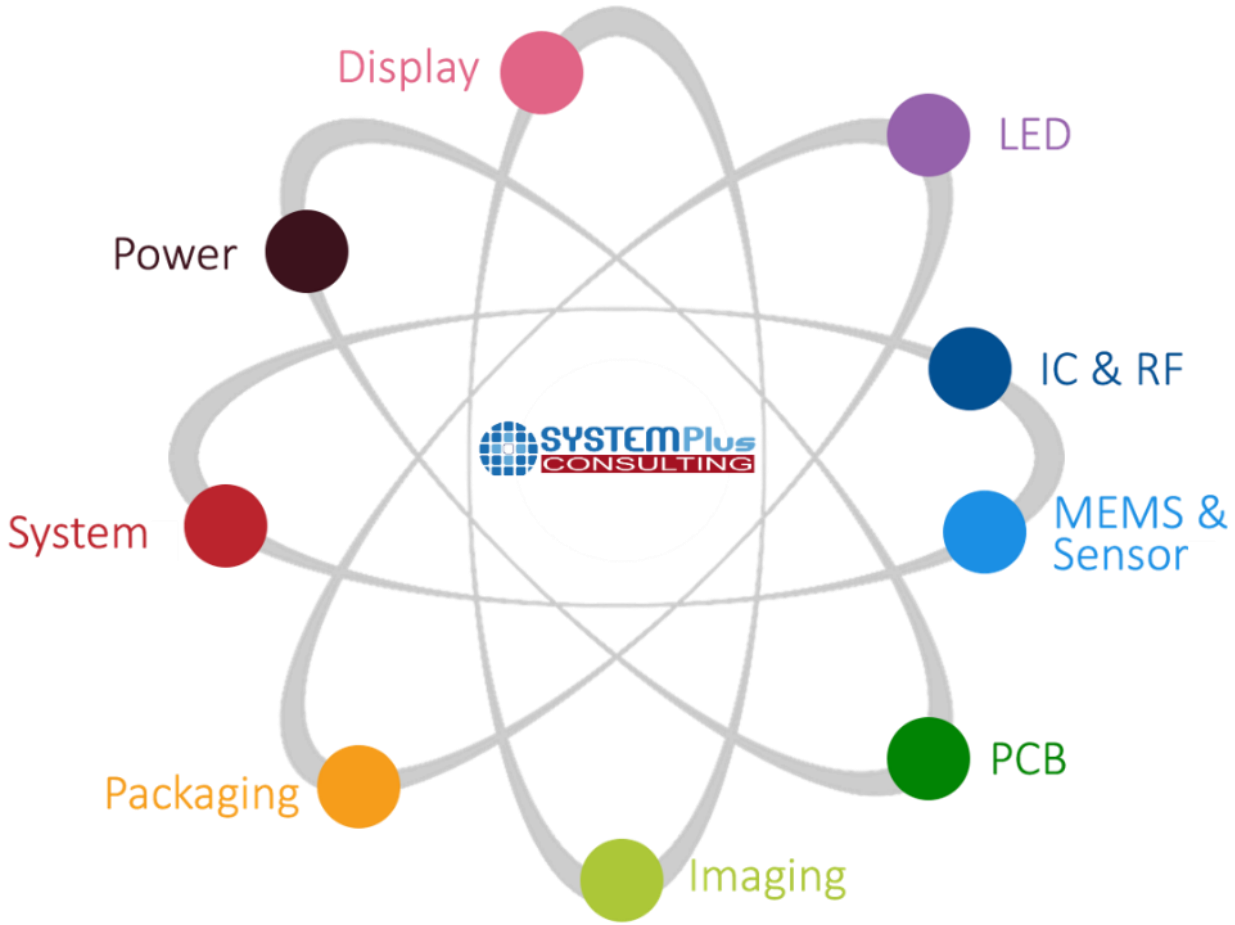
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Headquarters

21 rue La Noue Bras de Fer
44200 Nantes
FRANCE
+33 2 40 18 09 16
sales@systemplus.fr

Europe Sales Office

Lizzie LEVENEZ
Frankfurt am Main
GERMANY
+49 151 23 54 41 82
llevenez@systemplus.fr

America Sales Office

Steve LAFERRIERE
Phoenix
USA
(310) 600-8267
laferriere@yole.fr

Asia Sales Office

Takashi ONOZAWA
Tokyo
JAPAN
onozawa@yole.fr

Mavis WANG
GREATER CHINA
wang@yole.fr

www.systemplus.fr

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